

PRE-1970s NUCLEAR BOMBERS

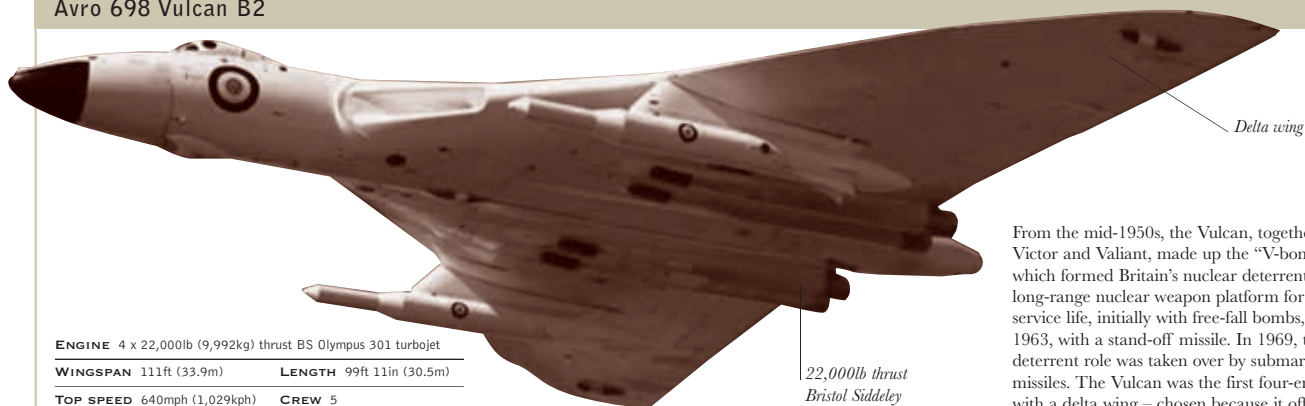
IN THE SECOND HALF OF THE 1940s, the first nuclear-bomber force in the United States consisted of aircraft conceived for conventional strategic bombing in World War II. Meanwhile, work started on devising a custom-built nuclear bomber. It was assumed that the best way to penetrate air defenses would be to fly as high and as fast as possible. For both these accomplishments, jet engines offered the best solution, but with the serious drawback that they could not at the time achieve the required range. It did not help that early nuclear bombs were large and heavy. Various ingenious solutions were tried involving rocket-assisted takeoff and combined jet and piston power. Eventually the range problem was resolved by improved jet engines and in-flight refueling. From 1957 the installation of SAMs, plus improvements in the performance of interceptors, led to a radical rethink. The bombers were forced to adopt a low-altitude attack profile, attempting to creep under radar. At the same time nuclear devices were getting smaller. By the end of the 1960s, strike aircraft such as the fighter-designated F-111 were preferred for low-level penetration, with ICBMs as the core of nuclear deterrence.

B-52 BOMBER

The Boeing B-52 had a relatively short life as the key delivery system for America's nuclear deterrent, but a long afterlife as a conventional bomber. See pages 286–87.



Avro 698 Vulcan B2



ENGINE 4 x 22,000lb (9,992kg) thrust BS Olympus 301 turbojet
WINGSPAN 111ft (33.9m) **LENGTH** 99ft 11in (30.5m)
TOP SPEED 640mph (1,029kph) **CREW** 5
ARMAMENT 21 x 1,000lb (454kg) bombs, nuclear bombs, or 1 x Blue Steel missile

22,000lb thrust Bristol Siddeley Olympus turbojet engine

Delta wing

From the mid-1950s, the Vulcan, together with the Victor and Valiant, made up the “V-bomber” force which formed Britain’s nuclear deterrent. It served as a long-range nuclear weapon platform for almost all its service life, initially with free-fall bombs, and then from 1963, with a stand-off missile. In 1969, the nuclear deterrent role was taken over by submarine-launched missiles. The Vulcan was the first four-engined aircraft with a delta wing – chosen because it offered a unique combination of good load-carrying capabilities, high subsonic speed at altitude, and long range.

Boeing B-47 Stratojet

One of the first jet bombers to serve with the US Strategic Air Command, the B-47 was classified as a medium bomber, even though its range and bombload would have made it a heavy bomber a few years earlier. The B-47, with air refueling and drop tanks, replaced the B-50 (the upgraded version of the B-29 Superfortress). Over 2,000 served from 1950 to 1966.

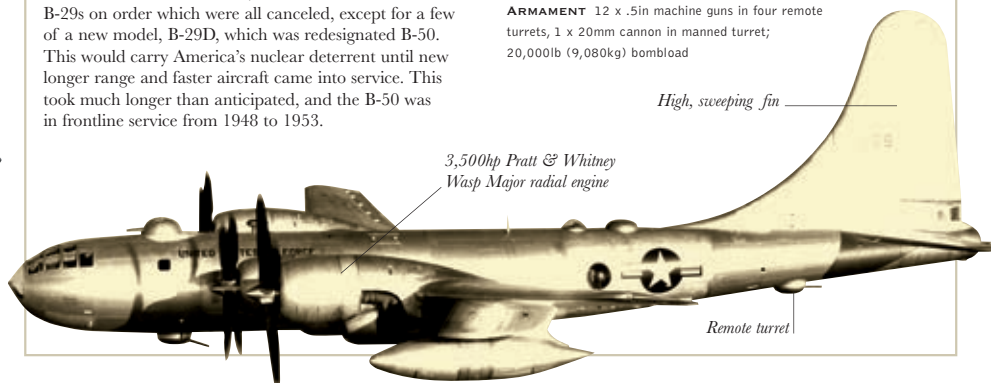


ENGINE 6 x 7,200lb (3,266kg) thrust GE J47-GE-25 turbojet
WINGSPAN 116ft (35.4m) **LENGTH** 107ft (32.6m)
TOP SPEED 606mph (980kph) **CREW** 3
ARMAMENT 2 x 20mm cannon; 20,000lb (9,080kg) bombload

Boeing B-50D

The most powerful and sophisticated bomber of WWII, the Boeing B-29 dropped thousands of tons of conventional bombs and two nuclear bombs on Japan. At the time of the surrender, there were thousands of B-29s on order which were all canceled, except for a few of a new model, B-29D, which was redesignated B-50. This would carry America’s nuclear deterrent until new longer range and faster aircraft came into service. This took much longer than anticipated, and the B-50 was in frontline service from 1948 to 1953.

ENGINE 4 x 3,500hp P&W R-4360-35 Wasp Major air-cooled radial
WINGSPAN 141ft 3in (43.1m) **LENGTH** 99ft (30.2m)
TOP SPEED 380mph (611kph) **CREW** 8
ARMAMENT 12 x .5in machine guns in four remote turrets, 1 x 20mm cannon in manned turret; 20,000lb (9,080kg) bombload



3,500hp Pratt & Whitney Wasp Major radial engine

High, sweeping fin

Remote turret